

Project 1 – Abnormal EV Charging Sessions

Task 1: Data Understanding, Audit and Preparation

Mestrado em Engenharia Informática
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Resumo

Este relatório descreve a Tarefa 1 do Projeto 1: compreender, auditar e preparar 441 077 sessões de carregamento de veículos elétricos, com o objetivo de prever, no momento em que cada sessão começa, se esta vai terminar de forma anormal. O ficheiro não traz o alvo. Derivámo-lo da causa de término e validámo-lo contra as colunas de histórico do próprio conjunto de dados, obtendo 18,6% de sessões anormais.

Excluímos as oito colunas que só existem depois da sessão e um carimbo temporal de disponibilidade duvidosa. Num diagnóstico, essas colunas fariam subir o PR-AUC de 0,604 para 0,834, um resultado impossível de reproduzir em produção. Os dados não têm duplicados nem registos inválidos, e os valores em falta são estruturais. O volume, a prevalência e o tipo de avaria mudam ao longo dos dois anos, pelo que a avaliação é cronológica, com um período inicial excluído e purga das sessões cujo resultado ainda não seria conhecido.

Retemos 29 variáveis em 7 grupos. Todas as operações que aprendem com os dados são ajustadas apenas no período de treino. O histórico recente de avarias do posto de carregamento é o sinal mais forte.

Abstract

This report covers Task 1 of Project 1: understanding, auditing and preparing 441,077 electric-vehicle charging sessions in order to predict, when each session starts, whether it will end abnormally. The file does not contain the target. We derived it from the end cause and validated it against the dataset's own history columns, which gives 18.6% abnormal sessions.

We excluded the eight columns that only exist after the session and one timestamp whose availability is doubtful. In a diagnostic, those columns would raise PR-AUC from 0.604 to 0.834, a result that could not be reproduced in production. The data contain no duplicates and no invalid records, and the missing values are structural. Volume, prevalence and fault type change over the two years, so evaluation is chronological, with an excluded burn-in period and a purge of sessions whose outcome would not yet be known.

We retain 29 features in 7 groups. Every operation that learns from the data is fitted on the training period only. Recent failure history at the charging post is the strongest signal.

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Acronyms

AUC	Area Under the Curve
BMS	Battery Management System
EV	Electric Vehicle
NaN	Not a Number (missing value)
PCA	Principal Component Analysis
PR-AUC	Area under the Precision–Recall curve
PSI	Population Stability Index
ROC-AUC	Area under the Receiver Operating Characteristic curve
SMOTE	Synthetic Minority Over-sampling Technique
VIF	Variance Inflation Factor

Chapter 1

Introduction

The goal is to predict, at the moment an EV charging session starts, whether it will end abnormally. Every preparation decision answers two questions: would this information exist at that moment, and is it trustworthy? The data are an educational adaptation of the China EV Charging Dataset [1]: 441,077 sessions and 47 columns, from 31 Dec 2019 to 31 Dec 2021, at 92 charging posts in 13 stations and 3 districts of Jiaxing. All numbers come from the notebook `TP1_Task1_Data_Understanding_and_Preparation.ipynb` (seed 42), and references of the form §x.y point to its sections.

1.1 Target definition

The file has no `is_Abnormal` column, so we derive it from `end_cause` (15 categories). The dataset’s pre-computed abnormal-count history columns were built from the true label, so we recomputed them under three candidate mappings (Table 1.1, §4.2). Only mapping A reproduces them. We adopt it, pending the instructor’s confirmation. The classes are about 4.4:1, so an “always normal” model reaches 81% accuracy, and accuracy is not an adequate metric.

Table 1.1: Candidate definitions of the target and agreement with the history columns

Definition of “abnormal”	Rows matching the supplied counts (6 windows)	Prevalence
A. 13 fault causes (“Charging ends normally” and “User stops charging” are normal)	$\geq 99.97\%$ in every window	18.58%
B. Everything except “Charging ends normally”	0.3–36%	53.93%
C. Charger faults only (vehicle-side faults count as normal)	2–70%	12.11%

Chapter 2

Data Audit and Exploration

2.1 Availability at prediction time

Seven post-hoc variables (energy, electricity cost, service charge, amount, actual payment, end time, payment time) and `end_cause` describe the result of the session, so we exclude them. On their own they reach univariate ROC-AUC 0.82–0.85, above any legitimate variable (best 0.77, Figure 2.1). Adding them to the model raises PR-AUC from 0.604 to 0.834 (§15.3), which is the optimistic, non-deployable result a leaky preparation would report. We also exclude `Order creation time`: it is stamped after the start in 56% of sessions and adds nothing measurable (−0.003 PR-AUC).

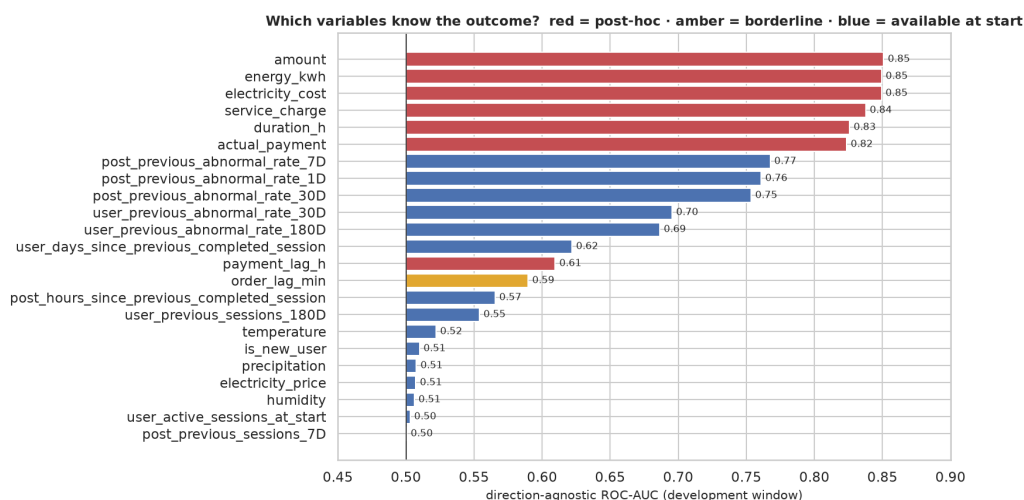


Figure 2.1: Univariate ROC-AUC of each variable, coloured by availability at the start of the session

2.2 History attributes

Meaning. Recomputing the history attributes from the timestamps reproduces the session counts in at least 99.96% of rows and the concurrency count in 99.97% (§4.3). Only sessions that ended before the current start are counted, so there is no look-ahead.

Operational availability. Post history needs a store that updates per-post counts when each session ends, and the platform already records end causes. User history also

needs the account to be identified at start. Our validation and test values assume that this store is updated continuously.

Privacy. User history is a behavioural profile tied to a pseudonymous identifier. Removing it costs 0.037 PR-AUC, against 0.195 for post history (§15.1), so the modelling task will compare models with and without it.

Generalisation and cold start. History replaces identities, so it works for accounts unseen in training (40% of validation and 61% of test sessions). New accounts (about 13% of sessions) get the prior rate 0.2 and missing time-since values, and the model ranks them worse (§15.2). Post cold start cannot be tested, because every validation and test post appears in training.

2.3 Data quality

We removed no record and investigated each anomaly before deciding (Table 2.1; §3, §8, §9).

Table 2.1: Data-quality checks, findings and decisions

Check	Finding	Decision
Duplicates	None (full rows or user + post + start)	No action
Formatting	Trailing tab in 2 timestamp columns and 18,370 location values	Stripped on load
Missing values	Only in the 4 “time since previous” columns; NaN equals the “no previous” flag in 100% of rows	Structural, not imputed with mean or median; informative (14.4% vs 19.3% abnormal)
Zero energy	54,621 sessions, 45,563 of them abnormal	Kept: early faults and cancellations deliver no energy
Post-hoc timestamps	Durations stop at 23.99 h; 87,202 payments recorded before the end	Not used; censoring noted as a limitation
Weather	One value per district and day	Treated as a forecast, as instructed (limitation)
Outliers	Idle posts (39% abnormal on return), fleet accounts, rainstorm days; Isolation Forest flags informative sessions	All kept; log and capping in the linear view only
Cardinality	92 posts, the smallest with 244 training sessions; fault causes from 20,114 down to 5 sessions	No grouping needed for a binary target

2.4 What relates to abnormal termination

We measure relationships on the development window only (Apr 2020 to Jun 2021). The post matters most among static variables: abnormal rates range from 4% to 61% across posts (Cramér’s V 0.30, against 0.23 for station and 0.11 for district), and post-level rates correlate at Spearman $\rho = 0.81$ between 2020 and 2021. Calendar and weather effects are

small ($V \leq 0.042$). Recent failure history is the strongest signal: the seven post-history variables top the univariate ranking (AUC 0.71–0.77) because faults cluster in time at the same post (§7.4). The supplied rates are smoothed as $(\text{abnormal} + 1) / (\text{sessions} + 5)$, so an entity without history gets 0.2. Sessions concentrate in a few accounts (the top 1% hold 34%), so `user_id` is not usable and user history replaces it.

2.5 Temporal stability

The process is not stationary (Figure 2.2). Monthly volume falls to 1,391 sessions in Feb 2020, coinciding with the COVID-19 restrictions, and reaches 25–28k in autumn 2021. Prevalence ranges from 13.7% to 26.3% per month, and faulty-connection faults fall from 28% of abnormal sessions in 2020Q1 to about 6% from 2020Q3. 90–92 posts are active each month until posts 17–20 stop on 28 Sep 2021, and active accounts grow from about 2,000 to 5,600–7,500 per month. The history attributes are censored at the start of the file: 31% of Jan 2020 sessions are flagged as new users (about 13% later), and their PSI is far above 0.25 in Q1-2020 and at most 0.10 from Q3-2020. A random split would mix these regimes and overstate performance, so evaluation is chronological.

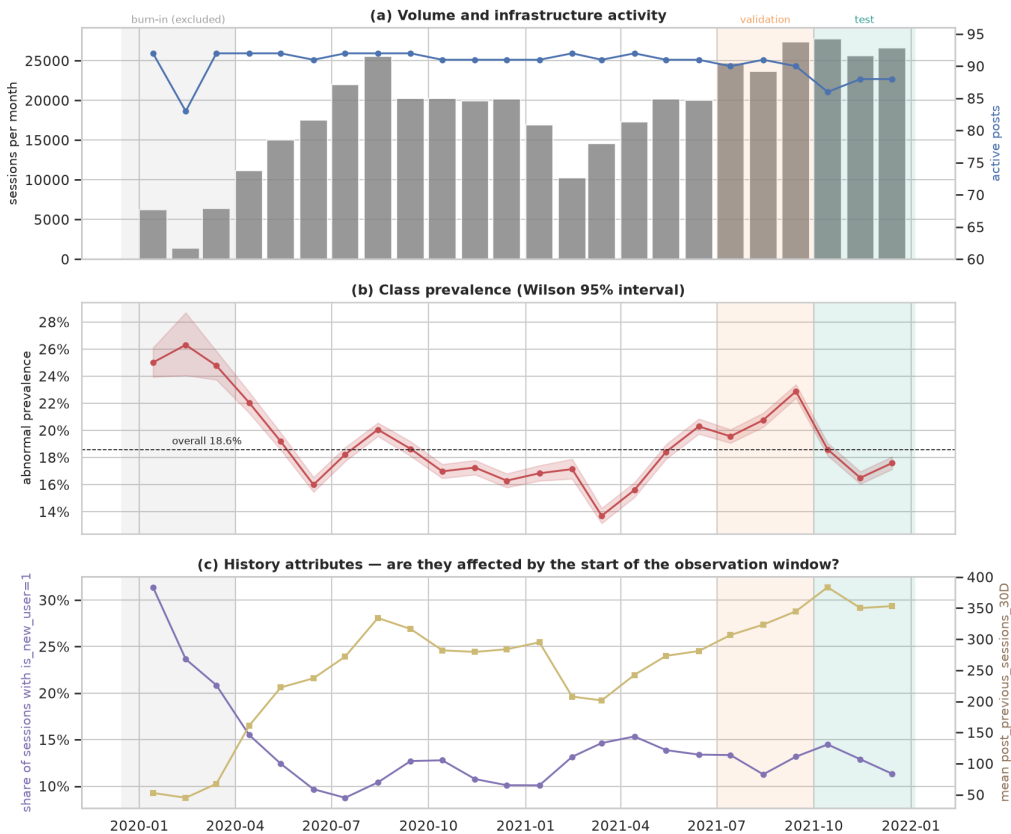


Figure 2.2: Monthly volume, abnormal rate and history availability

Chapter 3

Preparation and Evaluation

3.1 Evaluation design

Table 3.1: Chronological split

Period	Dates	Sessions	Abnormal
Burn-in (excluded)	31 Dec 2019 to 31 Mar 2020	14,017	25.0%
Train	1 Apr 2020 to 30 Jun 2021	271,182	17.8%
Purged	Train starts that end after 1 Jul 2021	28	n/a
Validation	Jul to Sep 2021	75,769	21.1%
Test (used once)	Oct to Dec 2021	80,081	17.6%

Table 3.1 shows the split. The burn-in is excluded because its history values are censored and its fault regime is obsolete; including it changes PR-AUC by only +0.0009 (§15.4). Training rows must also *end* before the cutoff, since their label is unknown until then (purge). Hyper-parameters will be tuned with three expanding-window folds inside the training period. We will report PR-AUC with lift over the base rate, ROC-AUC, the Brier score, and precision and recall at a threshold chosen on validation.

3.2 Retained features and learned operations

We retain 29 features in 7 groups: calendar (hour, day of week), tariff period, location (`post_id`, station, district), weather (4), post history (8), user history (10) and concurrency (1). We drop `electricity_price` (a function of the tariff), the six abnormal-count columns (exact functions of counts and rates; adding them back changes nothing, §15.4), the post “no history” flags (constant after the burn-in), `user_no_previous_completed_session` ($\phi \approx 1$ with `is_new_user`), `is_weekend` ($V = 0.001$) and `user_id`.

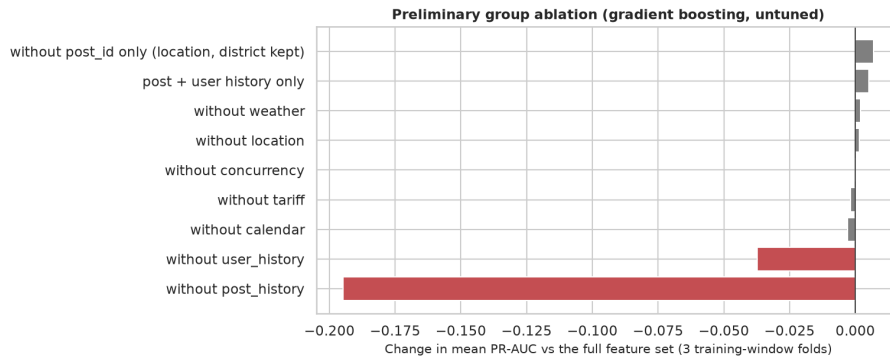
Every operation that learns from data is fitted on the training period only, inside scikit-learn pipelines (Table 3.2, §13). Fitting on all periods would change the parameters; for example, the structural-NaN fill would move from 394 to 511 days. `log1p` and the sine/cosine encoding of hour and weekday are stateless.

Table 3.2: Learned operations, all fitted on the training period

Operation	What is learned	Used by
Category codes / one-hot (<code>post_id</code> , station, district, tariff)	Category list	Tree / linear view
Winsorisation of heavy-tailed variables	0.1th and 99.9th percentiles	Linear view
Structural-NaN fill (“never happened”)	Maximum; indicators kept	Linear view (trees keep NaN)
Standardisation	Mean and standard deviation	Linear view
Reduced numeric set (19 variables, max VIF 65 to 8.6)	Correlations on a training sample	Unregularised linear models
Class-imbalance treatment	Class weights or resampling	Training folds only
Rare-category grouping, PCA	Nothing	Not needed / not adopted

3.3 Preliminary diagnostics

A fixed, untuned gradient-boosting model on the training-window folds reaches PR-AUC 0.604 (folds 0.569–0.640; lift 3.56) and ROC-AUC 0.832. These runs check preparation decisions; they are not model selection. Removing post history drops PR-AUC to 0.409, removing user history costs 0.037, and the other groups, `post_id` included, change it by at most 0.007 (Figure 3.1). Class weights leave the ranking unchanged (0.602 vs 0.604) but push the mean prediction to 0.34 against a prevalence of 0.17, so the prepared data are not resampled and SMOTE is not used. For new accounts, ROC-AUC is 0.714 (lift 2.38), against 0.845 (lift 3.80) for returning users.

**Figure 3.1:** Change in PR-AUC when each feature group is removed

Chapter 4

Synthesis

1. **Main characteristics and problems.** The target had to be derived and validated (18.6% abnormal). Eight outcome columns and one doubtful timestamp would leak it. Missing values are structural and no record is invalid. Volume, prevalence, fault mix and site mix drift, and history is censored in Q1-2020. Risk concentrates by post, and recent post failure history is the strongest signal.
2. **Evidence for the decisions.** The history-count oracle (target), AUC and PR-AUC inflation (leakage), the 100% match between NaN and its flag (missing values), the outlier investigation, PSI and monthly analyses (split), exact identities confirmed in §15.4 (redundancy), and the ledger of training-only parameters.
3. **Retained variables.** 29 features in 7 groups. Post history is essential and user history is useful; the context groups remain candidates for the modelling task.
4. **Limitations and assumptions.** The target mapping is inferred; weather is assumed to be a forecast but is daily; durations are censored at 24 h; payment and order timing semantics are unknown; the causes of drift are hypotheses; post cold start cannot be tested; history in validation and test assumes a continuously updated store; fleet accounts dominate user counts; the diagnostics use one untuned model.
5. **Implications for modelling.** Chronological split with purge, expanding-window cross-validation and a single use of the test set; PR-AUC with lift, ROC-AUC and calibration instead of accuracy; imbalance handled by the threshold or in-fold weights; tree ensembles and a regularised logistic-regression baseline; models compared with and without `post_id` and user history, with cold-start users reported separately.

Bibliography

- [1] Zhang, Y., Xu, T., Chen, T., et al. (2025). A high-resolution electric vehicle charging transaction dataset with multidimensional features in China. *Scientific Data*, 12, 643. <https://doi.org/10.1038/s41597-025-04982-1>